About NineScrolls LLC

U.S.-BASED PROCESS-EQUIPMENT SELECTION, CONFIGURATION, AND SUPPORT

NineScrolls LLC helps universities, national laboratories, and R&D and advanced-manufacturing teams across the United States select, configure, and support semiconductor process equipment. We start from the work — your materials, target process window, sample size, and facility conditions — and match the platform to it, rather than starting from a catalog.

As a U.S.-based partner, we coordinate selection, configuration, quoting, delivery, and after-sales support so a research team gets a process-ready platform and a local point of contact.

Process-first platform selection

We confirm materials, process window, sample size, and facility conditions first, then match the platform — not the other way around.

Configured around your lab

Platforms are specified to your throughput, wafer sizes, gases, and space, rather than sold as a fixed SKU.

U.S.-based project coordination and support

Selection, configuration, quoting, delivery, and after-sales support are coordinated locally.

Peer-reviewed validation for represented platforms

The corresponding platform classes appear in real peer-reviewed research — validating the process capability, not NineScrolls-owned equipment or NineScrolls-authored papers.

Peer-Reviewed Validation for the Platforms We Represent

Research using corresponding plasma, deposition, and vacuum process platforms has appeared in Nature Portfolio journals, Advanced Materials, Materials Today, and Scientific Reports.

Real published research using corresponding process platforms.

Nature Communications · 2021
Near-ideal van der Waals rectifiers based on all-two-dimensional Schottky junctions
Corresponding RIE process platform · 245 citations (as of Jul 2026)

Light: Science & Applications · 2026
On-chip nonlocal metasurface for color router
Corresponding RIE process platform

Advanced Materials · 2026
Diffraction-free omnidirectional antireflection binary metasurface
Corresponding ICP process platform

Materials Today · 2026
Solar-blind deep-UV photodetector based on β-Ga_2O_3/AlN/p-Si
Corresponding PECVD process platform · 9 citations (as of Jul 2026)

Scientific Reports · 2025
Experimental study of inductively coupled plasma etching of patterned single crystal diamonds
Corresponding ICP process platform · 4 citations (as of Jul 2026)

These publications validate represented platform classes and process capabilities. They are not claims of NineScrolls-branded installed-base citations.

EQUIPMENT PLATFORM

RIE Etcher Series

Reliable anisotropic plasma etching for university and R&D labs — dielectric patterning, polymer removal, surface activation, and device prototyping across silicon, compound, and 2D materials.

Broad material range. Si-based films, compounds (InP/GaN/GaAs), 2D materials, and metals — plus failure-analysis work — in one chamber.

Wide, repeatable process window. 300–1000 W RF, 4 gas lines, and a −70 to 200 °C stage; non-uniformity under ±5% (edge exclusion).

Tunable by design. Showerhead gas feed and a configurable discharge gap give tunable etch profiles.

Lab-ready and configurable. 1.0 × 1.0 m footprint; open-load or load-lock; cost- or performance-optimized builds.

SPECIFICATION	PARAMETERS
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Etching Materials	Si-Based (Si/SiO2/SiNx/SiC/Quartz etc.), Compounds (InP/GaN/GaAs/Ga2O3/ZnS etc.), 1D&2D Materials (MoS2/BN/Graphene etc.), Metals (Au/Pt/W/Ta/Mo etc.), Failure Analysis, etc.
Vacuum	TMP & Mechanical Pump
RF Power	Full Range 300-1000W, optional
Gas System	4 lines (Standard) or customized
Wafer Cooling	Water Cooling or He Backside Cooling optional
Wafer Stage Temperature Range	From -70°C to 200°C, optional
Non-Uniformity	Less than ±5% (Edge Exclusion)

TYPICAL APPLICATIONS

Semiconductor R&D · Dielectric patterning · Polymer removal · Surface activation

Explore configurations & request a quote →

EQUIPMENT PLATFORM

ICP Etcher Series

High-density plasma etching for demanding research — silicon, MEMS, diamond, and compound semiconductors, with independent control of plasma density and ion energy.

Independent plasma control. Separate source (1000–3000 W) and bias (300–1000 W) RF let plasma density and ion energy be tuned independently.

Hard-to-etch materials. Si, compounds (InP/GaN/GaAs), 2D materials, metals (W/Ta/Mo), and diamond in one platform.

Gentle when needed. Low-power plasma technology with an ion damage-free option for sensitive devices.

Wide process window. 5 gas lines, He backside cooling, and a −70 to 200 °C stage; non-uniformity under ±5% (edge exclusion).

SPECIFICATION	PARAMETERS
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Etching Materials	Si-Based ($Si/SiO_2/SiN_x/SiC$/Quartz etc.), Compounds ($InP/GaN/GaAs/Ga_2O_3$ etc.), 2D Materials (MoS_2/BN/Graphene etc.), Metals (W/Ta/Mo etc.), Diamond, Failure Analysis, etc.
Vacuum	TMP & Mechanical Pump
RF Power	Source 1000-3000W, Bias 300-1000W, optional
Gas System	5 lines (Standard) and He backside cooling, or customized
Wafer Stage Temperature Range	From -70°C to 200°C, optional
Non-Uniformity	Less than ±5% (Edge Exclusion)

TYPICAL APPLICATIONS

MEMS fabrication · Advanced packaging · Photonics · Power electronics

Explore configurations & request a quote →

EQUIPMENT PLATFORM

Plasma Photoresist Stripping Series

Dedicated plasma stripping and ashing for photoresist, post-etch residue, and organic contamination removal — gentle enough for damage-sensitive process flows.

Thorough organic removal. Strips photoresist, PMMA, and PS nanospheres, and cleans residue on 2D materials and failure-analysis samples.

Uniform, repeatable ashing. Chamber-center pump-down and uniform gas feed-in; non-uniformity under ±5% (edge exclusion).

Temperature to match the resist. Water-cooled stage adjustable from 5 to 200 °C, with 300–1000 W RF and 2 gas lines.

Compact and simple. 0.8 × 0.8 m footprint with open-load handling; cost- or performance-optimized builds.

SPECIFICATION	PARAMETERS
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Etching Materials	Organics (PR/PMMA/PS nanosphere etc.), 2D Materials (MoS_2/BN/Graphene etc.), Failure Analysis, etc.
Vacuum	Mechanical pump
RF Power	Full range 300-1000W, optional
Gas System	2 lines (Standard) or customized
Wafer Cooling	Water cooling
Wafer Stage Temperature Range	From 5°C to 200°C, optional
Non-Uniformity	Less than ±5% (Edge Exclusion)

TYPICAL APPLICATIONS

Photoresist stripping · Plasma ashing · Post-etch residue cleaning · Organic contamination removal

Explore configurations & request a quote →

EQUIPMENT PLATFORM

IBE/RIBE Series

Directional ion beam and reactive ion beam etching for difficult-to-etch research materials — magnetic films, noble metals, optical materials, and multilayer stacks.

Swappable ion sources. Kaufman (up to 6 inch) or RF (up to 12 inch) sources in an easy-to-swap design, matched to your materials.

Controlled beam geometry. Stage tilt from 0° to 90° and rotation from 1–10 rpm for tunable etch angles and profiles.

Clean, high-vacuum process. Base vacuum better than 7E-7 Torr, with a water-cooled 5 to 20 °C stage and optional He backside cooling.

Easy to run and maintain. 1.0 × 0.8 m footprint; open-load or load-lock; sample holder and ion source designed for easy operation.

SPECIFICATION	KAUFMAN ION SOURCE	RF ION SOURCE
Wafer Size Range	up to 6 inch	up to 12 inch
Gas System	1 line (standard) or customized	3 line (standard) or customized
Wafer Stage Motion	Tilt from 0° to 90°, Rotation from 1-10 rpm/min	
Wafer Stage Cooling	From 5 to 20°C, Water cooling; He backside cooling optional	
Vacuum	TMP & Mechanical Pump	
Base Vacuum	Better than 7E-7 Torr	
Non-Uniformity	Less than ±5% (Edge Exclusion)	

TYPICAL APPLICATIONS

- Magnetic materials
- Noble metal patterning
- Optical device fabrication
- MEMS / NEMS

Explore configurations & request a quote →

EQUIPMENT PLATFORM

ALD Series

Atomic-level thin-film deposition for research stacks — conformal coatings, high-k dielectrics, and passivation layers, even on demanding 3D structures.

Conformal on high-AR structures. Box-in-box chamber, multiple gas inlets, and vertical precursor throw for high-AR step coverage.

Broad material library. Oxides (Al2O3/HfO2/SiO2), nitrides (TiN/AlN/SiNx), and metals (Pt/Pd/W) with 2–6 precursor lines.

Tight uniformity. Non-uniformity under ±1% (Al2O3) across 4–12 inch wafers, supersize optional.

Thermal and plasma modes. Wafer temperatures from 20 to 400 °C plus an optional 300–1000 W remote plasma source.

SPECIFICATION	PARAMETERS
Wafer Size Range	4, 6, 8, 12 inch or Supersize optional
Growth Materials	Oxides (Al2O3/HfO2/SiO2/TiO2/Ga2O3/ZnO etc.), Nitrides (TiN/TaN/SiNx/AlN/GaN etc.), Metals (Pt/Pd/W etc.), etc.
Vacuum	TMP & Mechanical Pump
Base Vacuum	Better than 5E-5 Torr
RF Power	Remote Plasma 300-1000W, optional
Number of Precursor	2-6 lines or customized
Temperature of Source	From 20°C to 150°C (Standard), 200°C optional
Wafer Temperature Range	From 20°C to 400°C, higher temperature optional
Non-Uniformity	Less than ±1% (Al2O3)

TYPICAL APPLICATIONS

Gate dielectrics · Passivation layers · MEMS coatings · Energy storage materials

Explore configurations & request a quote →

EQUIPMENT PLATFORM

PECVD Series

Low-temperature plasma-enhanced CVD for research thin films — dielectric layers, passivation, optical coatings, MEMS stacks, and process development.

Dual-frequency RF for low stress. Electrode RF drive with dual-frequency options for film-stress tuning and process control.

Core dielectric films. α-Si:H, SiO2, SiNx, and SiC deposition across 4–12 inch wafers.

Wide process window. 500–2000 W RF, 6 gas lines, and a 20 to 400 °C stage; non-uniformity under ±5% (edge exclusion).

Tunable plasma geometry. Variable discharge gap, chamber liner, and electrode temperature control adapt to different processes.

SPECIFICATION	PARAMETERS
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Deposition Materials	Si-Based (α-Si:H/SiO2/SiNx/SiC etc.), etc.
Vacuum	Roots & Mechanical Pump
RF Power	Full Range 500-2000W, optional
Gas System	6 lines (Standard) or customized
Wafer Stage Temperature Range	From 20°C to 400°C, higher temperature optional
Non-Uniformity	Less than ±5% (Edge Exclusion)

TYPICAL APPLICATIONS

Passivation layers · Interlayer dielectrics · Optical coatings · MEMS membranes

Explore configurations & request a quote →

EQUIPMENT PLATFORM

HDP-CVD Series

High-density plasma CVD for dense dielectrics and void-free trench fill — STI, IMD, PMD, and advanced packaging process development.

Built for gap-fill. Step coverage tuned as a process parameter for trench-fill and dense dielectric work.

Independent source and bias. 1000–3000 W source and 300–1000 W bias RF, each configurable to the process.

Dielectric film set. Si, SiO2, SiNx, SiON, and SiC films with 6 gas lines and a 20 to 200 °C stage.

Lab-ready platform. 1.0 × 1.5 m footprint; process design kits; open-load or load-lock handling.

SPECIFICATION	PARAMETERS
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Deposition Materials	Si/SiO2/SiNx/SiON/SiC, etc.
Vacuum	TMP & Mechanical Pump
RF Power	Full Range: Source 1000-3000W, Bias 300-1000W, optional
Gas System	6 lines (Standard) or customized
Wafer Stage Temperature Range	From 20°C to 200°C
Non-Uniformity	Less than ±5% (Edge Exclusion)

TYPICAL APPLICATIONS

- STI gap-fill
- IMD / PMD dielectrics
- Advanced packaging dielectrics
- TSV isolation workflows

Explore configurations & request a quote →

EQUIPMENT PLATFORM

PVD Magnetron Sputtering Series

Configurable DC/RF magnetron sputtering for research thin films — metals, dielectrics, nitrides, oxides, and magnetic, optical, and compound materials.

Multi-target flexibility. 2–6 magnetron sources, face-down or face-up, with tiltable angle and tunable deposition distance.

DC and RF in one system. DC or RF power with an automatic switcher, plus RF substrate bias for in-situ clean and process tuning.

Cold to high-temperature substrates. From water cooling up to 400, 800, or 1200 °C options, with a rotational, temperature-controlled electrode.

Clean, repeatable films. Base pressure better than 5E-7 Torr, in-situ or independent-chamber pre-clean, and non-uniformity under ±5%.

SPECIFICATION	PARAMETERS
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Magnetron Sputtering Source	2-6 optional
Substrate Temperature	Water-cooling, 400°C, 800°C, 1200°C, optional
Gas System	2 lines (Standard), number of lines customized
Power	DC or RF customized, automatic switcher
Non-Uniformity	Less than ±5%
Pre-Cleaning	Independent chamber or in-situ, RF plasma, optional
Base Pressure	Better than 5E-7 Torr, higher vacuum customized

TYPICAL APPLICATIONS

Metal contacts · Magnetic films · Optical coatings · Compound semiconductors

Explore configurations & request a quote →

EQUIPMENT PLATFORM

Coater/Developer & Hotplate Series

Modular coat, develop, bake, HMDS, and EBR process control for repeatable photolithography — from small research pieces to pilot-line substrates.

Precise spin control. Coater to 8000 rpm and developer to 5000 rpm, each held to ±1 rpm.

Broad substrate range. Small pieces, 2–12 inch wafers, or square substrates across coater, developer, and hotplate modules.

Modular by design. Coater, developer, and hotplate module count customized, with options down to dispense systems and developer temperature.

Integrated bake. Hotplates up to 200 °C (higher optional) with 3 lift-pins, compatible down to 2 inch.

SPECIFICATION	COATER	DEVELOPER
Wafer Size Range	Small-piece, 2, 4, 6, 8, 12 inch or Square optional	
Max. Spin Speed	8000 rpm ±1rpm	5000 rpm ±1rpm
Max. Acceleration	8000 rpm/s	5000 rpm/s
Dispense Arm	Up to 2 photoresist lines	Up to 2 developer lines and deionized water line
Interlock	Vacuum pressure, uncover etc.	

HOTPLATE SPECIFICATIONS	
Wafer Size Range	Small-piece, 2, 4, 6, 8, 12 inch or Square optional
Max. Temperature	Up to 200°C, Higher Temperature optional
Lift-Pins	3 lift-pins, minimum compatible 2 inch

TYPICAL APPLICATIONS

- Photoresist coating
- HMDS priming
- Developer processing
- Lift-off preparation

Explore configurations & request a quote →

EQUIPMENT PLATFORM

Plasma Cleaner Systems

Benchtop plasma cleaning for surface activation, cleaning, and hydrophilic or hydrophobic treatment.

Multi-gas plasma processing. O_2, N_2, or Ar plasma at 0–300 W or 500 W RF with automatic matching and 2–3 gas lines.

Gentle on diverse materials. Contact-free treatment of photoresist, PMMA, PDMS, organics, and semiconductor, optical, and biomedical materials.

Batch-friendly. Single- or multi-wafer batches up to 6 inch, with MFC or manual flow control (0–300 sccm).

SPECIFICATION	PARAMETERS
Wafer Size Range	≤ 6 inch, multi-wafer batch processing
RF Power	0 ~ 300 W / 500 W, automatic matching
Gas System	2 ~ 3 gas lines
Process Gases	O_2, N_2, Ar
Flow Control Range	0 ~ 300 sccm
Flow Control	MFC or manual control
Pump System	Mechanical pump (TMP optional)
Operation	Touchscreen control, fully automated
Footprint	630 mm × 600 mm
Compatible Materials	Photoresist (PR); PMMA; PDMS; HMDS; organic films and polymers; semiconductor materials; optical materials; biomedical materials
Main Functions	Surface cleaning; surface activation; hydrophilic / hydrophobic treatment; functional group modification (–OH / –H / –COOH); contact-free plasma processing
Typical Applications	Chemical & biological laboratories; failure analysis; optical components; biomedical and medical devices

Family options: HY-4L · HY-20L · HY-20LRF · PLUTO-T · PLUTO-M · PLUTO-F

TYPICAL APPLICATIONS

Surface activation | Surface cleaning | Failure analysis | Optical & biomedical device prep

Explore configurations & request a quote →

EQUIPMENT PLATFORM

MEB-600 E-Beam Evaporation Series

Multi-source e-beam and thermal evaporation for optical and IR research — photonic crystals, optical multilayers, IR sensors, and lift-off metallization at research-grade purity.

Two evaporation sources. E-beam plus thermal resistance for metals, oxides, fluorides, and IR films.

Precise thickness control. In-situ QCM endpoint monitoring; uniformity ≤±5% within Φ6 in.

Optical and IR stacks ready. High-purity films for photonic crystals, optical multilayers, and IR sensors.

Flexible operation. Manual, semi-automatic, or full-automatic; suited to lift-off metallization and patterning.

SPECIFICATION	PARAMETERS
Substrate	Φ6 in x1 flat substrate holder
Substrate Heating	RT to 300°C
Substrate Rotation	1–10 rpm
E-Gun Crucible	6 pockets, 17 cc each
E-Beam Power	~10 kW
Uniformity	≤±5% within Φ6 in
Thickness Control	In-situ QCM endpoint
Vacuum	6.7×10^{-5} Pa ultimate vacuum
Operating Modes	Manual / semi-auto / full-auto
Sources	E-beam + thermal resistance
Materials	Metals, oxides, fluorides, IR films

TYPICAL APPLICATIONS

- Infrared image sensors
- Ge/ZnS photonic crystals
- UV down-conversion films
- Optical AR coatings

Explore configurations & request a quote →

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